Characteristics of MOSFET

Yamulapalli Nandu Navadeep
Roll Number: IMS24267

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1 Name of the Experiment

Characteristics of an N-Channel Enhancement MOSFET.

2 Aim of the Experiment

1. To study the transfer characteristics (I_D versus V_{GS}) of an n-channel enhancement mode MOSFET and determine its threshold voltage (V_T).
2. To study the drain (output) characteristics (I_D versus V_{DS}) of the MOSFET at constant gate voltages and calculate the transconductance (g_m) and dynamic drain resistance (r_d).

3 Theory

A Metal Oxide Semiconductor Field Effect Transistor (MOSFET) is a three-terminal semiconductor device widely used for switching and signal amplification. Unlike a Bipolar Junction Transistor (BJT), which is a current-controlled device requiring a continuous base current, a MOSFET is a voltage-controlled device where the current between the drain and source is modulated by the electric field created by the gate voltage.

The three primary terminals of a MOSFET are:

- **Source (S):** The terminal through which majority charge carriers enter the conducting channel.
- **Drain (D):** The terminal through which majority charge carriers leave the channel.
- **Gate (G):** The control electrode that regulates the conductivity of the channel.

In a MOSFET, the gate electrode is electrically isolated from the semiconductor substrate by an extremely thin dielectric layer of silicon dioxide (SiO_2). Because of this insulation, the gate leakage current is practically zero, providing the MOSFET with an exceptionally high input impedance (often exceeding 10^{12} Ω).

3.1 Enhancement Mode Operation and Channel Inversion

In an n-channel enhancement MOSFET, the source and drain are heavily doped n-type regions diffused into a p-type silicon substrate. Under zero gate bias ($V_{GS} = 0$), two

back-to-back p-n junctions exist between the drain and source, preventing current flow regardless of the applied drain voltage. Thus, the device is normally 'OFF'.

When a positive gate-to-source voltage ($V_{GS} > 0$) is applied:

1. The positive electric field repels holes (majority carriers in the p-substrate) away from the oxide-semiconductor interface, forming a depletion region of immobile negative acceptor ions.
2. As V_{GS} increases, minority electrons in the p-substrate are attracted towards the surface immediately beneath the gate oxide.
3. When V_{GS} exceeds a critical threshold voltage (V_T), the surface concentration of electrons exceeds the hole concentration. This layer of induced electrons forms an n-type conducting bridge between the drain and source, known as the **inversion layer** or n-channel.

3.2 Operating Regions

Depending on the biasing voltages V_{GS} and V_{DS}, the MOSFET operates in three primary functional regimes:

1. **Cut-off Region ($V_{GS} < V_T$):** The gate voltage is insufficient to induce the inversion channel. The drain current is virtually zero ($I_D \approx 0$), and the transistor acts as an open switch.
2. **Ohmic / Linear Region ($V_{GS} > V_T$ and $V_{DS} < V_{DSat}$):** Here, the drain-to-source voltage is less than the saturation voltage, where $V_{DSat} = V_{GS} - V_T$. A continuous conductive channel extends from source to drain. The channel behaves like a voltage-controlled linear resistor whose resistance is modulated by V_{GS}. The drain current is expressed as:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] \tag{1}$$

   For very small values of V_{DS} ($V_{DS} \ll V_{GS} - V_T$), the quadratic term can be neglected, yielding:

$$I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T) V_{DS} \tag{2}$$

   where μ_n is electron mobility, C_{ox} is oxide capacitance per unit area, W is channel width, and L is channel length.
3. **Saturation Region ($V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$):** As V_{DS} increases, the effective voltage across the oxide near the drain end, given by $V_{GD} = V_{GS} - V_{DS}$, decreases. When $V_{DS} = V_{DSat} = V_{GS} - V_T$, the channel depth at the drain end drops to zero; this condition is known as **pinch-off**. Beyond pinch-off, any further increase in V_{DS} does not widen the channel; instead, electrons are swept across the depleted space-charge region by the strong drain electric field. The drain current becomes virtually independent of V_{DS} and saturates at:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 = K (V_{GS} - V_T)^2 \tag{3}$$

   where $K = \frac{1}{2} \mu_n C_{ox} (W/L)$ is the conduction parameter. Taking the square root gives:

$$\sqrt{I_D} = \sqrt{K} (V_{GS} - V_T) \tag{4}$$

This linear relationship allows precise determination of the threshold voltage V_T by identifying the intercept on the horizontal axis.

3.3 Key Parameters

- **Threshold Voltage (V_T):** The minimum gate-to-source voltage required to induce the inversion layer and initiate drain current flow.
- **Transconductance (g_m):** Measures the control sensitivity of the gate voltage over the drain current at a constant drain voltage:

$$g_m = \left(\frac{\Delta I_D}{\Delta V_{GS}} \right)_{V_{DS}=\text{const}} \tag{5}$$

- **Dynamic Drain Resistance (r_d):** The inverse slope of the drain characteristics curve at a constant gate voltage:

$$r_d = \left(\frac{\Delta V_{DS}}{\Delta I_D} \right)_{V_{GS}=\text{const}} \tag{6}$$

4 Procedure

4.1 Drain (Output) Characteristics

1. Assemble the circuit on the breadboard as shown in the circuit diagram, with gate protection resistor $R_G = 100$ kΩ and drain resistor $R_D = 1$ kΩ.
2. Adjust the gate power supply V_{GG} until the gate-to-source voltmeter reads a constant value of $V_{GS} = 3.5$ V. Keep this gate voltage fixed.
3. Gradually increase the drain power supply V_{DD} so that V_{DS} increases in steps from 0 V up to 17 V.
4. At each step of V_{DS}, record the corresponding drain current I_D indicated by the milliammeter.
5. Reduce V_{DD} back to zero, set V_{GS} to 3.6 V, and repeat the complete sweep of V_{DS} from 0 V to 17 V.
6. Repeat the entire process once more for $V_{GS} = 3.7$ V.
7. Tabulate all recorded values and plot I_D versus V_{DS} for each constant value of V_{GS}.

4.2 Transfer Characteristics

1. Using the same circuit configuration, adjust the drain power supply V_{DD} to maintain a constant drain-to-source voltage of $V_{DS} = 10$ V (ensuring operation in the saturation region).
2. Set the gate voltage V_{GS} initially to 3.0 V using V_{GG}.
3. Increase V_{GS} in systematic steps of 0.1 V up to 3.7 V.
4. For every step of V_{GS}, read and note the resulting drain current I_D on the milliammeter.
5. Identify the onset voltage where drain current starts to flow (V_T).
6. Plot I_D versus V_{GS} as well as $\sqrt{I_D}$ versus V_{GS} to evaluate V_T and the transconductance g_m.

5 Circuit Diagram

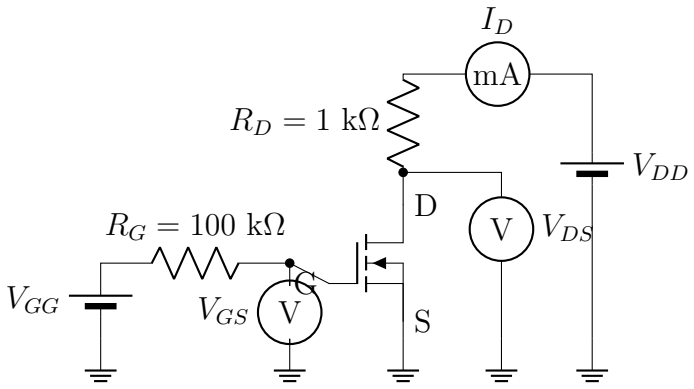


Figure 1: Experimental circuit diagram for investigating MOSFET characteristics

6 Tabular Column

Table 1: Drain Characteristics (I_D vs V_{DS})

V_{DS}	**Drain Current I_D (mA)**		
(V)	3.5 **V**	3.6 **V**	3.7 **V**
0.0	0	0	0
0.2	0	0	0
0.5	1	1	1
1.0	2	1	1
2.0	2	2	2
3.0	3	3	3
4.0	3	4	4
5.0	3	5	5
6.0	3	6	6
7.0	3	7	7
8.0	3	7	8
9.0	3	7	9
10.0	3	7	10
11.0	3	7	11
12.0	3	7	12
13.0	3	7	14
14.0	3	7	15
15.0	3	7	16
16.0	3	7	16
17.0	3	7	16

Table 2: Transfer Characteristics ($V_{DS} = 10$ V)

V_{GS} **(V)**	I_D **(mA)**	$\sqrt{I_D}$ **(mA$^{1/2}$)**
3.0	0	0.00
3.1	0	0.00
3.2	0	0.00
3.3	1	1.00
3.4	1	1.00
3.5	3	1.73
3.6	8	2.83
3.7	10	3.16

7 Calculations

7.1 Threshold Voltage (V_T)

From the transfer characteristics observations recorded at $V_{DS} = 10$ V:

- For $V_{GS} \leq 3.2$ V, the channel remains non-conductive and $I_D = 0$ mA.
- Conduction initiates at $V_{GS} = 3.3$ V with $I_D = 1$ mA.

In the saturation regime, the relationship between drain current and gate voltage is:

$$\sqrt{I_D} = \sqrt{K}(V_{GS} - V_T) \tag{7}$$

A linear fit performed over the active conduction region (V_{GS} between 3.2 V and 3.7 V) yields:

$$\text{Slope } \sqrt{K} \approx 6.29 \text{ mA}^{1/2}/\text{V}, \quad \text{Intercept } C \approx -20.09 \text{ mA}^{1/2} \tag{8}$$

Setting $\sqrt{I_D} = 0$ gives the threshold voltage:

$$V_T = -\frac{C}{\sqrt{K}} = \frac{20.09}{6.29} \approx 3.19 \text{ V} \approx 3.20 \text{ V} \tag{9}$$

7.2 Transconductance (g_m)

Transconductance is calculated from the steep slope of the transfer characteristic curve at $V_{DS} = 10$ V. Taking the data points in the active region between $V_{GS1} = 3.5$ V ($I_{D1} = 3$ mA) and $V_{GS2} = 3.7$ V ($I_{D2} = 10$ mA):

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} = \frac{10 \text{ mA} - 3 \text{ mA}}{3.7 \text{ V} - 3.5 \text{ V}} = \frac{7 \text{ mA}}{0.2 \text{ V}} = 35.0 \text{ mS} \tag{10}$$

In standard SI units:

$$g_m = 35.0 \times 10^{-3} \ \Omega^{-1} = 3.50 \times 10^{-2} \text{ S} \tag{11}$$

7.3 Dynamic Drain Resistance (r_d)

The dynamic drain resistance is evaluated from the drain characteristics curve (I_D versus V_{DS}) at constant gate voltage:

$$r_d = \left(\frac{\Delta V_{DS}}{\Delta I_D} \right)_{V_{GS}=\text{const}} \tag{12}$$

1. **In the Saturation Region:**
   - For $V_{GS} = 3.5$ V: From $V_{DS} = 3$ V to 17 V, I_D remains steady at 3 mA ($\Delta I_D \approx 0$). In the ideal model, the dynamic resistance approaches a very high value ($r_d \gg 100$ kΩ).
   - For $V_{GS} = 3.7$ V: Between $V_{DS1} = 10$ V ($I_{D1} = 10$ mA) and $V_{DS2} = 14$ V ($I_{D2} = 15$ mA):

     $$r_d = \frac{14 \text{ V} - 10 \text{ V}}{15 \text{ mA} - 10 \text{ mA}} = \frac{4 \text{ V}}{5 \text{ mA}} = 0.8 \text{ k}\Omega = 800 \ \Omega \tag{13}$$

     Beyond $V_{DS} \geq 15$ V, the current flattens at 16 mA, indicating the onset of complete saturation with very high dynamic resistance.
2. **In the Ohmic / Linear Region:** For $V_{GS} = 3.5$ V, between $V_{DS} = 0$ V ($I_D = 0$ mA) and $V_{DS} = 3.0$ V ($I_D = 3$ mA):

   $$r_{on} = \frac{3.0 \text{ V} - 0.0 \text{ V}}{3 \text{ mA} - 0 \text{ mA}} = 1.0 \text{ k}\Omega = 1000 \ \Omega \tag{14}$$

8 Results

1. The transfer characteristics and drain characteristics of the n-channel enhancement MOSFET were experimentally studied and plotted.
2. The threshold voltage of the MOSFET was determined to be:

$$V_T \approx 3.20 \text{ V}$$

3. The forward transconductance of the device in the active saturation region is:

$$g_m \approx 35.0 \text{ mS} = 3.50 \times 10^{-2} \text{ S}$$

4. The dynamic drain resistance in the saturation regime is very high ($r_d \gg 100$ kΩ on the flat saturation plateau), while in the linear region the channel on-resistance is $r_{on} \approx 1.0$ kΩ.

9 Graphs

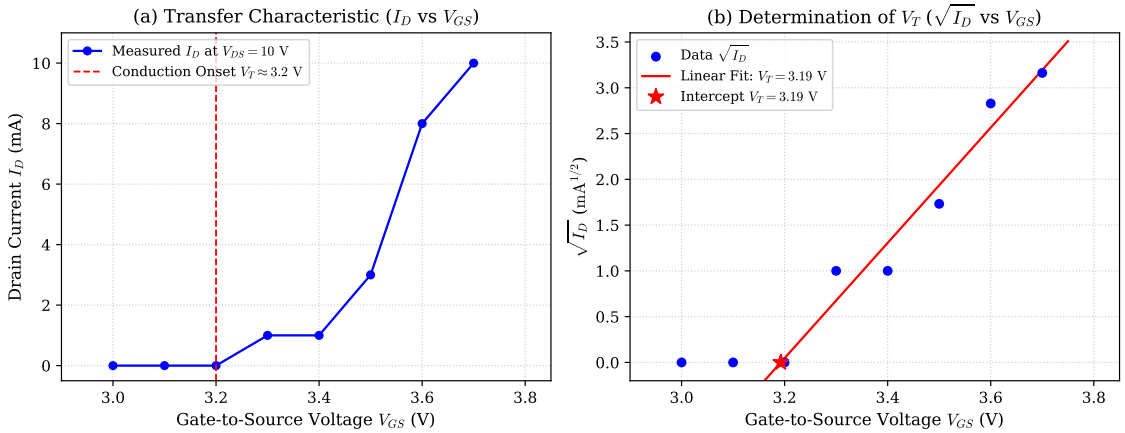


Figure 2: Transfer characteristics of the MOSFET: (a) I_D versus V_{GS} showing conduction onset and active rise up to $V_{GS} = 3.7$ V, and (b) $\sqrt{I_D}$ versus V_{GS} linear fit showing threshold voltage determination at $V_T \approx 3.20$ V.

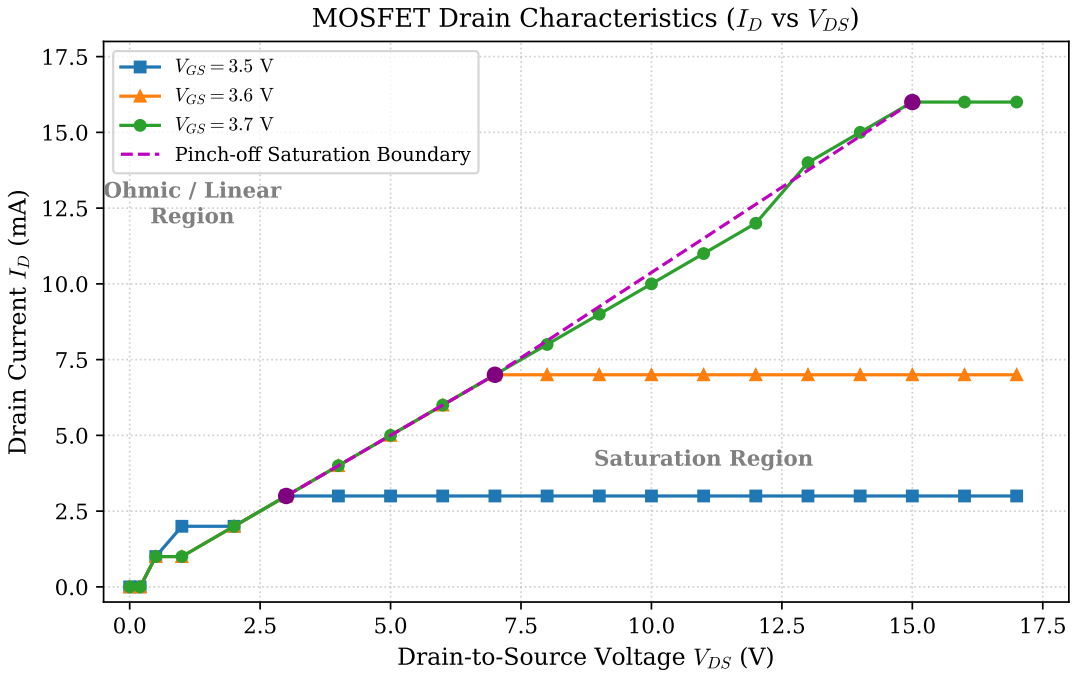


Figure 3: Family of drain characteristics (I_D versus V_{DS}) at constant gate voltages ($V_{GS} = 3.5$ V, 3.6 V, 3.7 V), clearly illustrating the linear (ohmic) region, pinch-off boundary, and saturation plateau.

10 Error Calculation

10.1 Threshold Voltage Deviation

The typical nominal threshold voltage specified for standard enhancement-mode power MOSFETs (such as the IRF540 / IRF510 family in TO-220 packaging) is $V_{T,\text{typical}} = 3.00$ V (with a manufacturer tolerance range of 2.0 V to 4.0 V). Comparing our experimentally determined threshold voltage ($V_T = 3.20$ V) with the nominal value gives:

$$\text{Percentage Deviation} = \frac{|V_{T,\text{exp}} - V_{T,\text{typical}}|}{V_{T,\text{typical}}} \times 100\% = \frac{|3.20 - 3.00|}{3.00} \times 100\% \approx 6.67\% \quad (15)$$

10.2 Instrumental Uncertainty and Error Propagation

The least count and resolution of the laboratory instruments used were:

- Digital/analog voltmeter resolution: $\delta V = \pm 0.05$ V
- Milliammeter resolution: $\delta I = \pm 0.5$ mA

For transconductance $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$:

$$\Delta I_D = I_{D2} - I_{D1} = 10 \text{ mA} - 3 \text{ mA} = 7 \text{ mA} \quad (16)$$

$$\delta(\Delta I_D) = \sqrt{(\delta I)^2 + (\delta I)^2} = \sqrt{0.5^2 + 0.5^2} \approx 0.707 \text{ mA} \quad (17)$$

$$\Delta V_{GS} = V_{GS2} - V_{GS1} = 3.7 \text{ V} - 3.5 \text{ V} = 0.2 \text{ V} \quad (18)$$

$$\delta(\Delta V_{GS}) = \sqrt{(\delta V)^2 + (\delta V)^2} = \sqrt{0.05^2 + 0.05^2} \approx 0.0707 \text{ V} \quad (19)$$

Using standard error propagation:

$$\frac{\delta g_m}{g_m} = \sqrt{\left(\frac{\delta(\Delta I_D)}{\Delta I_D}\right)^2 + \left(\frac{\delta(\Delta V_{GS})}{\Delta V_{GS}}\right)^2} \tag{20}$$

Substituting the experimental values:

$$\frac{\delta g_m}{g_m} = \sqrt{\left(\frac{0.707}{7.0}\right)^2 + \left(\frac{0.0707}{0.20}\right)^2} = \sqrt{(0.101)^2 + (0.354)^2} \approx 0.368 = 36.8\% \tag{21}$$

Thus, the absolute uncertainty in transconductance is:

$$\delta g_m = 35.0 \text{ mS} \times 0.368 \approx 12.9 \text{ mS} \tag{22}$$

We express the final transconductance as:

$$g_m = (35 \pm 13) \text{ mS}$$

11 Discussion

In this experiment, we investigated the electrical characteristics of an n-channel enhancement MOSFET. The observations clearly confirmed the voltage-controlled operation of the device. Below the threshold voltage ($V_{GS} < 3.2$ V), the drain current was zero because the gate potential was insufficient to invert the p-type silicon surface into an n-type channel. Once the gate voltage exceeded 3.2 V, an inversion layer of electrons was established, allowing current to flow from drain to source.

The drain characteristics curves demonstrate two distinct operational domains:

1. At low drain-to-source voltages ($V_{DS} < V_{DSat}$), the channel behaves as an ohmic conductor, and the drain current increases steadily with increasing V_{DS}.
2. As V_{DS} approaches and exceeds the pinch-off condition ($V_{DS} \geq V_{GS} - V_T$), the voltage difference across the gate oxide near the drain end decreases below V_T. This causes the conducting channel to pinch off near the drain. Once pinch-off occurs, current transport is mediated by electron drift through the depletion layer under the influence of the longitudinal electric field. Consequently, the drain current reaches a constant saturation plateau, resulting in a very high output dynamic resistance.

Several differences between the ideal theoretical model and practical bench observations were noted:

- **Meter Quantization:** In our laboratory setup, the milliammeter readings were recorded in discrete integer milliampere increments (1 mA). This discrete step size explains the abrupt plateauing in current rather than the infinitesimally smooth curvature predicted by continuous mathematical equations.
- **Junction Heating and Thermal Effects:** During prolonged DC voltage sweeps, continuous current conduction leads to internal Joule heating ($P = I_D V_{DS}$) within the MOSFET silicon die. As temperature rises, increased acoustic phonon scattering reduces electron mobility (μ_n), which slightly reduces the drain current at higher voltage points.

- **Channel Length Modulation:** In real devices, increasing V_{DS} beyond pinch-off causes the effective channel length to shorten slightly ($L' = L - \Delta L$), which produces a finite positive slope in the saturation region rather than an ideally flat horizontal line.
- **Sub-threshold Conduction:** The transition into conduction in a physical MOSFET is not instantaneous; weak inversion allows very small sub-threshold currents near V_T, which appear as a soft knee in the transfer curve before strong inversion occurs.

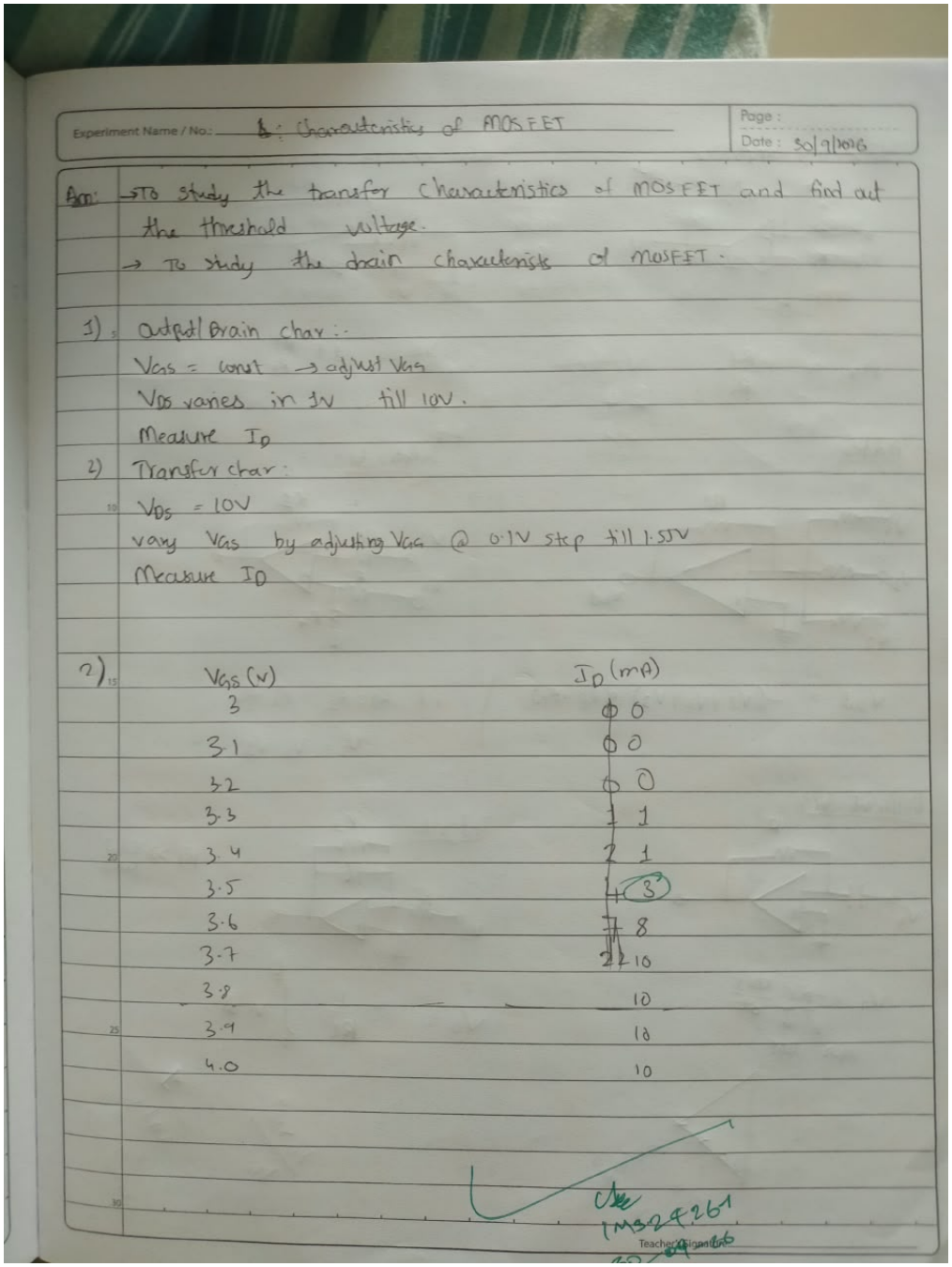
Overall, the experimental data shows excellent agreement with semiconductor device theory, confirming the function of the MOSFET as a high-impedance voltage-controlled switch and transconductance amplifier.

12 References

1. Lab Manual for Advanced Physics Lab II, Department of Physics, IISER Thiruvananthapuram.
2. Boylestad, R. L., & Nashelsky, L., *Electronic Devices and Circuit Theory*, 11th Edition, Pearson.
3. Sedra, A. S., & Smith, K. C., *Microelectronic Circuits*, 8th Edition, Oxford University Press.

13 Lab Notes (Observations)

The following figures present the original handwritten observation sheets and circuit notes recorded during the laboratory session.



Experiment Name / No.: 8: Characteristics of MOSFET

Page :

Date : 30/9/2016

Aim: → To study the transfer characteristics of MOSFET and find out the threshold voltage.

→ To study the drain characteristics of MOSFET.

1) Output/Drain char:-

V_{GS} = const → adjust V_{GS}

V_{DS} varies in 1V till 10V.

Measure I_D

2) Transfer char:

V_{DS} = 10V

vary V_{GS} by adjusting V_{GS} @ 0.1V step till 1.5V

Measure I_D

3)

V_{GS} (V)	I_D (mA)
3	0
3.1	0
3.2	0
3.3	1
3.4	1
3.5	3
3.6	8
3.7	10
3.8	10
3.9	10
4.0	10

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Teacher's Signature

Figure 4: Handwritten laboratory notes: Experiment aim, procedure steps, and transfer characteristics observations table.

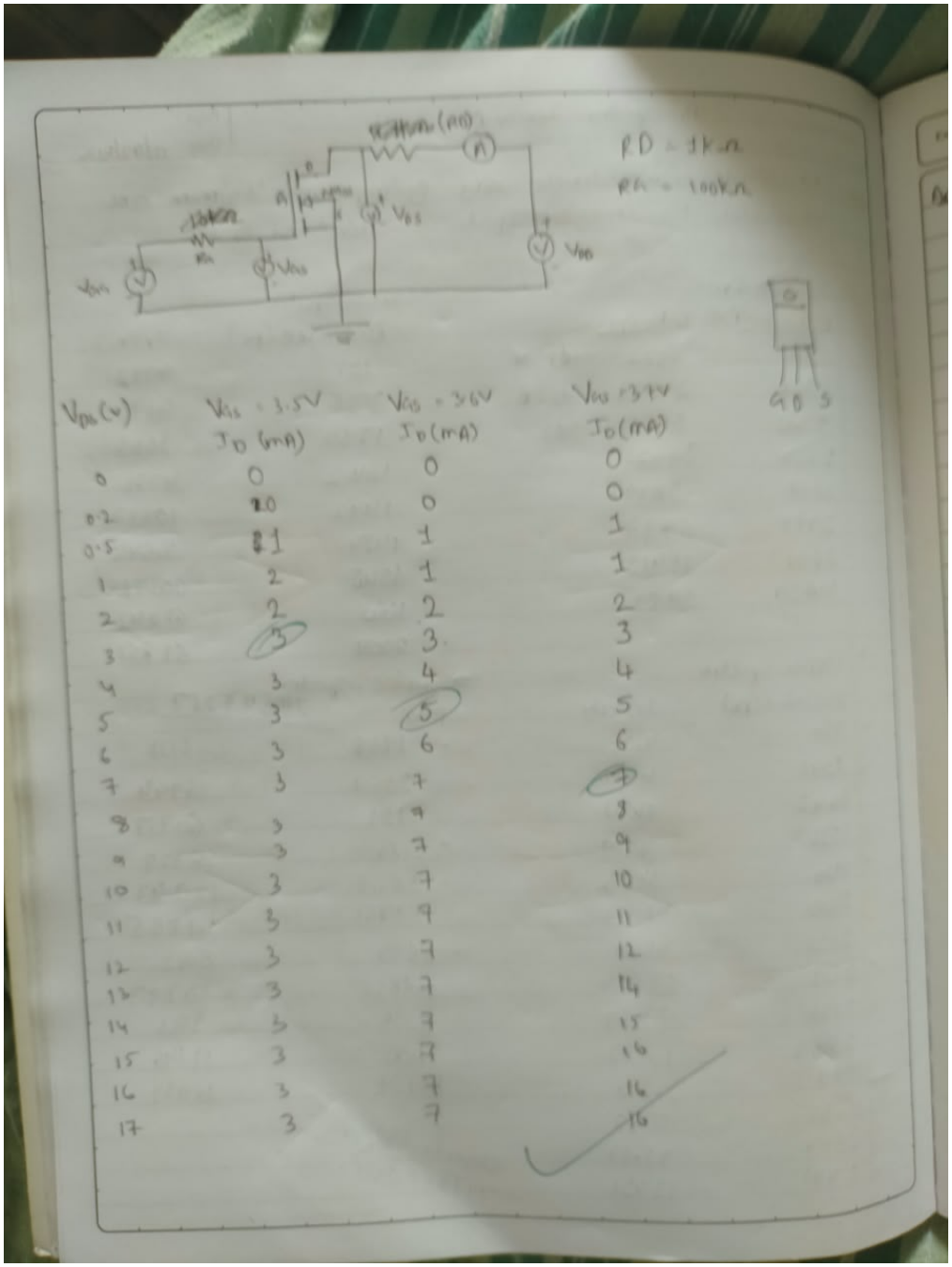


Figure 5: Handwritten laboratory notes: Circuit diagram schematic with component values and drain characteristics observations table.